

Param Nikhil Shah

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Education

Sardar Patel Institute of Technology (S.P.I.T.)

Bachelors of Computer Engineering

Minor in Internet of Things

Aug'23 – Present

Mumbai, India

GPA: 8.21

Projects

Financial Intelligence System | Python, XGBoost, DoWhy, AutoGen, FastAPI, Monte Carlo Feb '26 - Present

- Engineered a **causal, decision-time behavioral bias detection engine**, the only system that intervenes at the **exact moment** an investor decides, not retrospectively. Used **NOTEARS causal structure learning** and **DoWhy do-calculus** to quantify how much each bias causally drives a sell decision, achieving **85.7 % Macro F1** with **per-prediction SHAP attributions**, filling a gap no existing robo-advisor addresses.
- Engineered a **personalised long-horizon Monte Carlo simulator** (10,000 paths × 30 years) producing individual HOLD vs SELL wealth projections with P10/P50/P90 uncertainty bands, personalised to age, risk tolerance, and financial obligations
- Architected a **decision-centric Explainable AI layer** using a **multi-agent adversarial debate system(AutoGen)** that replaces static SHAP/LIME explanations with counterfactual narratives grounded in Monte Carlo simulation numbers paired with a **blockchain-style SHA-256 cryptographic audit trail** making every AI decision independently verifiable and tamper-evident.

CareerPath360.AI | React.js, Tailwind, Node.js, MongoDB, Python, NLP, REST APIs Aug'25 – Dec'25

- Designed and developed a **Python-based ML microservice** using **FastAPI** and **REST APIs** for resume parsing, ATS scoring and job matching, integrated with resume builder workflow.
- Used **NLP with Sentence Transformers (BERT)** and **feature engineering** for skill extraction and measuring semantic similarity between resumes and job descriptions.
- Trained and integrated an **XGBoost** model with a **hybrid ML + rule-based scoring pipeline**, for generating realistic ATS scores with **robust fallback mechanisms**.
- Incorporated **Google Gemini API** for context aware, AI-driven career guidance, resume improvement suggestions and intelligent job analysis and matching.

P2P Rental MarketPlace | JavaScript, React.js, MySQL, Node.js, Express.js April'25

- Developed a full-stack peer-to-peer rental platform enabling users to list, discover, and rent items using **React.js**, **MySQL**, and **Node.js**, enabling users to discover, rent and share items.
- Integrated **Google's Gemini AI API** to develop an intelligent in-app assistant that answers user queries about rentals, listings, policies, and booking help via natural language prompts.
- Implemented user trust mechanisms such as **verified profiles, reputation scores, and secure in-app chat** to enhance transparency and reduce friction in rentals.

Cryptocurrency Prediction and Analysis | Python, NumPy, Pandas, Scikit-learn, Tkinter Oct'24 – Feb'25

- Performed **Exploratory Data Analysis (EDA)** using **NumPy** and **Pandas**, and engineered features like open-close, low-high, and is-quarter-end to improve model accuracy.
- Trained a **Random Forest Classifier** using **Scikit-learn**, and evaluated it with **F1 Score, Confusion Matrix, and ROC-AUC curves**.
- Created a Python-based GUI using **Tkinter** with real-time prediction and 10+ interactive data visualizations using **Matplotlib and Seaborn**.

Skills

Languages: Java, Python, JavaScript, C, HTML, CSS

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, REST API, Tailwind CSS, NumPy, Pandas

Databases & Tools: MySQL, MongoDB, PostgreSQL, Git, GitHub

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Agile Development, Database Management Systems, NLP, Machine Learning

Certifications and Achievements

SCOPE - Certification by JP Morgan Chase in Agile methodology and Cloud.

Aug'24 – Present

JEE - Secured a rank of 33.1k across India in JEE-Mains and qualified for JEE Advanced.

Jun'23

Positions Of Responsibility

S.P.I.T. Placement Office - Training & Placement Coordinator

Dec'23 – Present

- Served as the primary Point of Contact between recruiters and students, managing campus drives and pre-placement sessions and analyzing placement data to identify target companies.

S.P.I.T. Sports Committee - Head of Marketing and Tournament Admin

Sep'23 – Aug'25

- Led marketing, PR and event operations for large-scale sports events, including a Half Marathon (1,500+ participants) and Spoorthi Sports Fest (5,000+ participants), ensuring seamless execution and strong brand visibility.